## Supplement S1: Benchmark

**Table S1. Benchmark evaluation categories and analytical significance.**

| Evaluation Category | Significance for the Analysis |
|---|---|
| Representation | Assesses the format in which guideline knowledge is represented, e.g., text-based, structured, computer-interpretable, or executable. |
| Decision Logic | Assesses the extent to which rules, conditions, threshold values, decision trees, or DMN-adjacent structures are mapped. |
| Process Logic | Assesses the extent to which sequences, workflows, roles, triggers, temporal aspects, or BPMN-adjacent structures are represented. |
| Evidence Reference | Assesses the extent to which evidence bases, strengths of recommendation, GRADE, Evidence-to-Decision frameworks, risk of bias, or clinical study links are mapped. |
| Interoperability | Assesses the connectivity to standards and systems such as FHIR, HL7, CQL, CDS, Electronic Health Records (EHR), or medical terminologies. |
| Updatability | Assesses the extent to which dynamic updates, Living Guidelines, or the continuous updates of evidence and recommendations are supported. |
| Implementation Closeness | Assesses how close the approach is to a practical implementation in CDSS, EHR systems, apps, registries, or other digital tools. |

To classify the identified approaches, a benchmark was established. It evaluates the approaches along seven criteria: degree of representation, decision logic, process logic, evidence reference, interoperability, updatability, and implementation closeness (Table S1). The matrix is not intended to provide a definitive evaluation of individual systems, but rather to present a comparative overview of their strengths and limitations with regard to machine-readable decision and process modelling.

The evaluation was conducted qualitatively using the ratings “++”, “+”, and “0”. Evaluation rules were defined in advance for each benchmark criterion to support a transparent and reproducible assessment (Table S2). A rating was assigned only if the respective contribution was identifiable in the title, abstract, full text, or described methodology of the investigated approach.

Evaluation rules were defined in advance for each benchmark criterion. A rating was assigned only if the respective contribution was identifiable in the title, abstract, full text, or the described methodology of the investigated approach.

"++" indicates a core contribution to the respective criterion. This rating was assigned if the criterion is a major focus of the approach, is methodologically elaborated, or plays a central role in the outcome of the paper.

"+" indicates a partial contribution to the respective criterion. This rating was assigned if the criterion is addressed but is not the primary focus, or is only relevant to a specific sub-aspect.
"0" indicates no identifiable focus. This rating was assigned if the criterion is not addressed or is only addressed indirectly within the paper.

**Table S2. Operationalization of benchmark rating criteria.**

| Criterion | Rating "++" | Rating "+" | Rating "0" |
|---|---|---|---|
| Degree of Representation | The approach explicitly describes a structured, computer-interpretable, or executable representation of guidelines, recommendations, or evidence. | The approach utilizes structured or digital elements without providing a complete representation model itself. | There is no identifiable formal or structured representation of guideline knowledge. |
| Decision Logic | Conditions, rules, decision points, decision tables, decision trees, or recommendation outputs are explicitly modeled. | Decision support or recommendations are utilized, but the underlying decision logic is only partially described. | Decision logic is not an identifiable component of the approach. |
| Process Logic | Sequences, workflows, roles, triggers, temporal dependencies, or process steps are explicitly modeled. | Process or workflow aspects are mentioned or partially considered, but are not systematically modeled. | Process logic is not an identifiable focus. |
| Evidence Reference | Evidence, clinical studies, GRADE, Evidence-to-Decision, Risk of Bias, evidence syntheses, or provenance are structured and mapped. | Evidence is mentioned or used as a foundation, but is not systematically or machine-readably modeled. | An evidence reference is not identifiable or plays no role in the approach. |
| Interoperability | The approach explicitly uses or addresses standards such as FHIR, CQL, openEHR, OMOP, RDF/OWL, terminologies, APIs, or EHR integration. | Interoperability is partially addressed, for instance, through data export, terminologies, or connectivity to individual systems. | There is no identifiable contribution to technical or semantic interoperability. |
| Updatability | Versioning, Living Guidelines, dynamic updates, continuous evidence updates, or update mechanisms are explicitly addressed. | Updating is mentioned or indirectly possible, but is not methodologically elaborated. | Updatability is not an identifiable component. |
| Implementation Closeness | The approach has been implemented or evaluated as a CDSS, app, registry, engine, dashboard, prototype, or clinical system. | The approach has an identifiable practical relevance but remains conceptual, prototypical, or hasn't been clinically evaluated. | There is no identifiable implementation or practical relevance. |

The resulting comparative benchmark of the selected approaches is presented in Table S3.

**Table S3. Comparative benchmark of selected approaches for digital and computable clinical guideline representation.**

| Authors | Title | Key Contribution | Representation | Decision Logic | Process Logic | Evidence Linkage | Interoperability | Updatability | Implementation Readiness | Main Gap |
|---|---|---|---|---|---|---|---|---|---|---|
| Nothacker et al. (2025) | Digitalisierung des AWMF-Leitlinienregisters mit FHIR, APIs und Nutzergruppen-Evaluation [1] | Digitalization of the AWMF guideline registry using FHIR, APIs, and user-group evaluation | ++ | + | + | + | ++ | + | ++ | Focuses on registry/conte infrastructure comprehensiv and process l |
| Scott et al. (2023) | Modelling clinical narrative as computable knowledge: The NICE computable implementation guidance project [2] | Translation of narrative NICE recommendations into a technology-agnostic logical model | ++ | ++ | ++ | + | + | + | ++ | Strong interm but not direct |
| Matson-Koffman et al. (2023) | An Integrated Process for Co-Developing and Implementing Written and Computable Clinical Practice Guidelines [3] | Parallel development of narrative and computable guidelines in 12 phases | ++ | ++ | ++ | + | ++ | + | ++ | Strong proces not a standalo modeling form |
| Lichtner et al. (2023) | Representation of evidence-based clinical practice guideline recommendations on FHIR [4] | FHIR-based representation of evidence-based guideline recommendations | ++ | ++ | + | ++ | ++ | + | + | Strong for recommenda but limited ex |
| HL7 International (2023) | Clinical Practice Guidelines on FHIR (CPG-on-FHIR) Implementation Guide (version 2.0.0) [5] | Computable guideline representation using FHIR knowledge artifacts | ++ | ++ | ++ | + | ++ | + | ++ | Requires sign knowledge en local impleme |
| HL7 International (2023) | Quality Measure (QM) / Clinical Quality Measure (CQM) Implementation Guide [5] | FHIR/CQL-based quality measurement | + | ++ | + | + | ++ | + | ++ | Strong for me not guideline |
| HL7 International (2023) | Evidence-Based Medicine on FHIR (EBMonFHIR) Implementation Guide [6] | Standard for evidence objects and recommendations | ++ | + | 0 | ++ | ++ | ++ | + | Evidence-focu than process |

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|---|---|---|---|---|---|---|---|---|---|---|
| Soares et al. (2024) | Making Science Computable Using Evidence-Based Medicine on Fast Healthcare Interoperability Resources: Standards Development Project [7] | Standardization of computable scientific evidence using FHIR | ++ | + | 0 | ++ | ++ | ++ | + | Evidence-cen logic indirect. |
| Seneviratne et al. (2023) | Semantically enabling clinical decision support recommendations [8] | Ontologies and provenance for explainable CDS | ++ | + | 0 | ++ | ++ | + | + | Strong seman explainability, process mode |
| Fortmann et al. (2022) | System for Context-Specific Visualization of Clinical Practice Guidelines (GuLiNav): Concept and Software Implementation [9] | Context-specific visualization of guidelines with BPMN and rules | ++ | + | ++ | 0 | + | 0 | + | Strong BPMN evidence stru |
| Garber & Patkar (2023) | Computer-interpretable guidelines: electronic tools to enhance the utility of thyroid nodule clinical practice guidelines and risk stratification tools [10] | Computer-interpretable guideline CDS for thyroid nodules | ++ | ++ | + | + | + | + | ++ | Domain-spec |
| Halms et al. (2024) | The impact of a digital guideline version on schizophrenia guideline knowledge: results from a multicenter cluster-randomized controlled trial [11] | RCT evaluating digital vs. PDF guidelines | + | 0 | 0 | + | 0 | + | ++ | Implementati not a computa formalism. |
| Börm (2021) | Leitlinienbasierter Clinical Decision Support – Anforderungen an evidenzbasierte Entscheidungsunterstützungssysteme [12] | Requirements for evidence-based CDS systems | + | ++ | + | ++ | + | + | + | Conceptual ra formal repres |
| Bazhenova et al. (2019) | From BPMN process models to DMN decision model [13] | Derivation of DMN models from BPMN processes | + | ++ | ++ | 0 | 0 | 0 | + | Methodologic not medical-s |
| Panos et al. (2024) | CAREPATH CDS Services [14] | Transformation of guidelines into CDS services | ++ | ++ | + | + | ++ | + | ++ | Project-specif architecture. |
| Michalowski et al. (2021) | Using graph rewriting to operationalize medical knowledge for the revision of concurrently applied clinical practice guidelines [15] | Graph-rewriting for operationalizing medical knowledge | + | ++ | + | 0 | + | 0 | + | Strong formal standardizatio |
| Martínez-Salvador et al. (2023) | A model-driven transformation approach for the modelling of processes in clinical practice guidelines [16] | Model-driven transformation for process modeling | ++ | + | ++ | 0 | + | 0 | + | Strong proces weaker evide |

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|---|---|---|---|---|---|---|---|---|---|---|
| Alonso-Coello et al. (2016) | GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well-informed healthcare choices [17] | GRADE Evidence-to-Decision framework | ++ | + | 0 | ++ | + | + | ++ | No execution |
| Elliott et al. (2017) | Living Systematic Reviews: 4 [18] | Living systematic reviews | + | 0 | 0 | ++ | + | ++ | + | Supports upd process mod |
| Garg et al. (2023) | CRIS: A type system for representing clinical recommendations [19] | CRTS type system for recommendations | ++ | ++ | 0 | + | + | 0 | + | Strong recom structure, no |
| Badra et al. (2023) | EdHibou / Kasimir: Ontology-based interface for medical decision support [20] | Ontology-based CDS with OWL reasoning | ++ | ++ | 0 | 0 | + | 0 | + | Older and les integrated. |
| Kaes et al. (2014) | The NNN Formalization: Review and Development of Guideline Specification in the Care Domain [21] | Formalization of nursing knowledge | ++ | + | ++ | + | + | 0 | + | Relevant to c processes, ol architecture. |
| Rahmaniheris et al. (2016) | Model-Driven Design of Clinical Guidance Systems using UPPAAL [22] | Model-driven clinical guidance systems | ++ | ++ | ++ | 0 | + | 0 | ++ | Strong verific specific. |
| Hatsek & Shahar (2021) | A Methodology for Bi-Directional Knowledge-Based Assessment of Compliance to Continuous Application of Clinical Guidelines [23] | Compliance assessment using Asbru | ++ | ++ | ++ | + | + | + | ++ | Strong for ad assessment. |
| Kogan et al. (2024) | A Hybrid Execution Environment for Computer-Interpretable Guidelines in PROforma [24] | Hybrid execution environment for PROforma | ++ | ++ | ++ | + | ++ | + | ++ | Dependent o ecosystem. |
| Kabak et al. (2025) | FHIR-RAG-MEDS: Integrating HL7 FHIR with Retrieval-Augmented Large Language Models for Enhanced Medical Decision Support [25] | FHIR-RAG-MEDS | ++ | ++ | + | ++ | ++ | + | ++ | RAG does no formal logic. |
| Shahar et al. (2026) | Integrating retrospective quality assessment with real-time guideline application to support the episodic application of clinical guidelines over significant time periods [26] | Retrospective quality assessment with real-time CDS | ++ | ++ | ++ | + | + | + | ++ | Not primarily representatio |
| Domingues (2026) | A hybrid decision support system using rule-based and AI methods: the OnCATs knowledge-based framework [27] | Hybrid rule-based and AI decision support | + | ++ | + | + | + | 0 | ++ | Partial guideli |
| Hsu et al. (2025) | Extracting Clinical Guideline Information Using Two Large Language Models: Evaluation Study [28] | LLM-based extraction of guideline information | ++ | ++ | 0 | + | + | + | + | Extraction is modeling. |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Luo et al. (2025) | A Computational Framework for Tailored Preventive Care Recommendations Using Electronic Health Records [29] | Tailored preventive care recommendations from EHRs | + | ++ | + | + | + | + | + | Less focus or transparency. |
| Bottrighi et al. (2025) | Ontology-based student testing through clinical guidelines: An AI approach [30] | Ontology-based guideline reasoning | ++ | ++ | + | 0 | + | 0 | + | Educational/t context. |
| Sitasuwan et al. (2025) | Factors Influencing Health Workers' Acceptance of Guideline-Based Clinical Decision Support Systems for Preventive Services in Thailand: Questionnaire-Based Study [31] | Acceptance factors for CDS systems | 0 | 0 | 0 | 0 | 0 | 0 | ++ | Implementati only. |
| Delourme et al. (2024) | Leveraging Guideline-Based Clinical Decision Support Systems with Large Language Models: A Case Study with Breast Cancer [32] | LLM-enhanced guideline-based CDS | + | ++ | + | + | + | 0 | ++ | Application ca than standard |
| Jammal et al. (2025) | Impact on clinical guideline adherence of Orient-COVID, a clinical decision support system based on dynamic decision trees for COVID19 management: A randomized simulation trial with medical trainees [33] | Dynamic decision-tree CDS | ++ | ++ | ++ | + | + | + | ++ | Less express BPMN/PROfo |
| Shahar et al. (2024) | Supporting the Episodic Application of Clinical Guidelines over Significant Time Periods [34] | Episodic application of guidelines | ++ | ++ | ++ | + | + | + | + | Generalizabil assessed. |
| Ebben et al. (2024) | Development and Implementation of Clinical Decision Algorithms for Oncology [35] | Narrative guideline to CDS operationalization | ++ | ++ | + | + | + | 0 | ++ | Formal depth implementati |
| Tignanelli et al. (2024) | A pragmatic, stepped-wedge, hybrid type II trial of interoperable clinical decision support to improve venous thromboembolism prophylaxis for patients with traumatic brain injury [36] | Interoperable CDS implementation | + | ++ | + | + | ++ | 0 | ++ | Domain-spec case. |
| Neumann et al. (2024) | TraumaFlow-development of a workflow-based clinical decision support system for the management of severe trauma cases [37] | Workflow-based trauma CDS | ++ | + | ++ | + | + | 0 | ++ | Domain-spec focus. |
| Berkhout et al. (2024) | Decision discovery using clinical decision support system decision log data for supporting the nurse decision-making process [38] | Decision discovery from CDS logs | + | ++ | + | 0 | + | 0 | + | Not a guidelir representatio |

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|---|---|---|---|---|---|---|---|---|---|---|
| Li et al. (2023) | The Application of Computer Technology to Clinical Practice Guideline Implementation: A Scoping Review [39] | Scoping review of technology for guideline implementation | + | + | + | + | + | + | + | Background/r |
| Beynon et al. (2023) | Digitalizing Clinical Guidelines: Experiences in the Development of Clinical Decision Support Algorithms for Management of Childhood Illness in Resource-Constrained Settings [40] | Digitalizing clinical guidelines | ++ | ++ | ++ | + | + | + | ++ | Transferability DMN/BPMN r evaluation. |
| Samal et al. (2023) | Refining Clinical Phenotypes to Improve Clinical Decision Support and Reduce Alert Fatigue: A Feasibility Study [41] | Computable phenotypes for CDS | + | + | 0 | + | + | + | + | Focus on elig |
| Richesson et al. (2020) | Measuring implementation feasibility of clinical decision support alerts for clinical practice recommendations [42] | Feasibility of CDS alerts | + | ++ | + | + | + | 0 | ++ | Useful for gap only. |
| Iglesias et al. (2020) | Comprehensive analysis of rule formalisms to represent clinical guidelines: Selection criteria and case study on antibiotic clinical guidelines [43] | Analysis of rule formalisms | ++ | ++ | + | 0 | + | 0 | + | FHIR integrat limited. |
| Bilici et al. (2019) | Concurrent Execution of Multiple Computer-interpretable Clinical Practice Guidelines and Their Interrelations [44] | Concurrent execution of multiple guidelines | ++ | ++ | ++ | 0 | + | 0 | + | Complex imp |
| Hendriks et al. (2019) | Transformation of the National Breast Cancer Guideline Into Data-Driven Clinical Decision Trees [45] | Guideline to decision-tree transformation | ++ | ++ | + | + | + | + | ++ | Less express workflows tha |
| Jafarpour et al. (2019) | Execution-time integration of clinical practice guidelines to provide decision support for comorbid conditions [46] | Runtime integration of multiple guidelines | ++ | ++ | ++ | + | + | 0 | ++ | Focuses on in rather than in transformatio |
| Quesada-Martínez et al. (2018) | Towards the semantic enrichment of Computer Interpretable Guidelines [47] | Semantic enrichment of CIGs | ++ | + | 0 | + | ++ | 0 | + | Complementa BPMN/DMN. |
| Peleg et al. (2003) | Comparing computer-interpretable guideline models: a case-study approach [48] | Comparison of CIG models | ++ | ++ | ++ | + | + | 0 | + | Foundational modern FHIR |

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